EDS 223: Homework Assignment 3

West Coast Marine Aquaculture

Karolina Sienko

2026-11-02

1. Load in libraries

```
library(tidyverse)
library(sf)
library(terra) # raster handling
library(tmap) # map making
library(viridisLite)
```

2. Load in and prepare data

```
# Load sea surface temperature data
sst_2008 <- rast("data/average_annual_sst_2008.tif")
sst_2009 <- rast("data/average_annual_sst_2009.tif")
sst_2010 <- rast("data/average_annual_sst_2010.tif")
sst_2011 <- rast("data/average_annual_sst_2011.tif")
sst_2012 <- rast("data/average_annual_sst_2012.tif")

# Load ocean depth data
depth <- rast("data/depth.tif")

# Load EEZ data
eez <- st_read("data/wc_regions_clean.shp")

# Combining SST rasters into a raster stack
sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
```

Function!!!

```
suitable_areas <- function(
  min_sst,
  max_sst,
  min_depth,
  max_depth,
  species_name
) {
  ## Process SST and depth data so that they can be combined
  # Average SST from 2008-2012 and convert from C to Kelvin
  avg_sst_C <- mean(sst_stack, na.rm = TRUE) - 273.15

  # Resample depth to match res and extent of SST using bilinear method approach
  depth_resampled <- resample(depth, avg_sst_C, method = 'bilinear')

  # Checking and converting depth to match SST resolution, extent, and CRS
  if (all(res(avg_sst_C) == res(depth_resampled))) {
    print("The resolutions match!")
  } else {
    warning("Update resolutions to match!")
    res(depth_resampled) <- res(avg_sst_C)
  }

  if (ext(avg_sst_C) == ext(depth_resampled)) {
    print("The extents match!")
  } else {
    warning("Update extents to match!")
    ext(depth_resampled) <- ext(avg_sst_C)
  }

  if (crs(depth_resampled) == crs(avg_sst_C)) {
    print("The coordinate reference systems match!")
  } else {
    warning("Update coordinate reference systems to match!")
    crs(depth_resampled) <- crs(avg_sst_C)
    if (crs(depth_resampled) == crs(avg_sst_C)) {
      print("The coordinate reference systems match!")}
  }

  ## Find suitable locations
```

```r
# Reclassifying SST and depth data into suitable locations
# SST reclassification matrix
rcl_sst <- matrix(
  c(
    -Inf, min_sst, 0,
    min_sst, max_sst, 1,
    max_sst, Inf, 0
  ),
  ncol = 3,
  byrow = TRUE
)

# Reclassifying sst
sst_zones <- classify(avg_sst_C, rcl = rcl_sst)

# Depth reclassification matrix
rcl_depth <- matrix(
  c(
    -Inf, max_depth, 0, # should max_depth be negative?
    max_depth, min_depth, 1,
    min_depth, Inf, 0
  ),
  ncol = 3,
  byrow = TRUE
)

# Reclassifying depth
depth_zones <- classify(depth_resampled, rcl = rcl_depth)

# Finding locations that satisfy both SST and depth conditions
# Adding oyster zone rasters
sst_depth_zones <- c(sst_zones, depth_zones)

# Finding locations that satisfy both SST and depth conditions
suitable_locs <- lapp(sst_depth_zones, function(sst, depth) sst * depth)

## Determine the total suitable area within each EEZ
# Making EEZ crs equal oyster_suitability CRS
sf::st_crs(eez) <- terra::crs(suitable_locs)

# Converting sf object to terra SpatVector
eez_vect <- terra::vect(eez)
```

```r
  # Rasterizing EEZ data
  eez_raster <- rasterize(eez_vect, suitable_locs, field = 'rgn_id')

  # Calculating area of each cell
  cell_sizes <- cellSize(suitable_locs, unit = "km", mask = TRUE)

  # Making sure only cells classified as 1 contribute to area totals
  suitable_area <- mask(cell_sizes, suitable_locs, maskvalues = 0)

  # Selecting suitable raster cells that fall within each EEZs region
  suitable_area_by_eez <- zonal(
    suitable_area,
    eez_raster,
    fun = 'sum',
    na.rm = TRUE
  )

  ## Mapping EEZ regions colored by amount of suitable area
  # Converting zonal results to a data frame
  suitable_area_df <- as_tibble(suitable_area_by_eez)

  # Joining suitable area to EEZ polygons
  eez_map <- left_join(eez, suitable_area_df, by = "rgn_id")

  # Creating map
  tm_shape(eez_map) +
    tm_fill(
      fill = "area",
      fill.legend = tm_legend(title = "Suitable Area (km^2)")
    ) +
    tmap_options(component.autoscale = FALSE) +
    tm_borders(lwd = .8) +
    tm_title(paste("Suitable", species_name, "Habitats\nby West Coast EEZ"), size = 1)
}
```

Optimal growth conditions for oysters & lobsters

Oysters: 11-30 deg C and 0-70 m below sea level

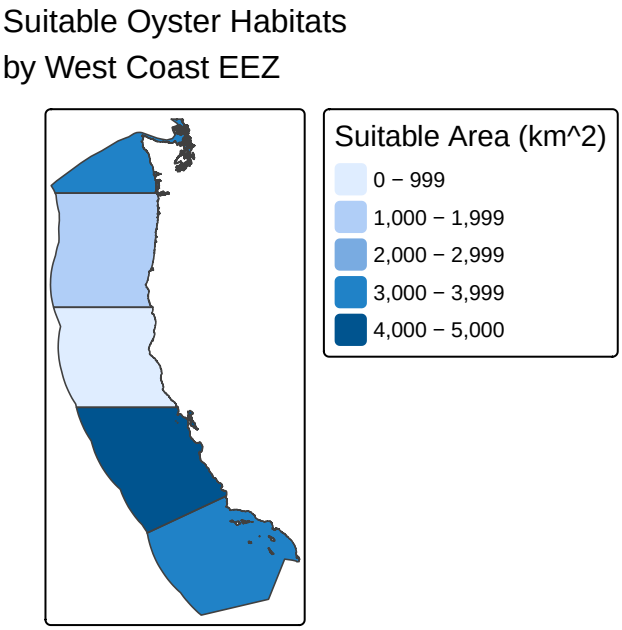
California Spiny Lobster: 10-21 deg C and depth: 0-150 m below sea level

Testing the function:

```
suitable_areas(11, 30, 0, -70, 'Oyster')
```

```
[1] "The resolutions match!"
[1] "The extents match!"
```

```
[1] "The coordinate reference systems match!"
```




```
suitable_areas(10, 21, 0, -150, 'Lobster')
```

```
[1] "The resolutions match!"
[1] "The extents match!"
```

```
[1] "The coordinate reference systems match!"
```

Suitable Lobster Habitats
by West Coast EEZ

